# Action Driver Closure Package

| Validation Memo – TRM TRR Lack of Proactive Obsolescence Governance – AD AB6665 | |
|---|---|
| **Action Driver Information** | |
| **Legal Entity:** | Banco Santander International (BSI) – ServiceNow company: SPB Miami |
| **Division:** | BSI IT – IT Operations / Infrastructure (ITOPS) |
| **Action Driver ID:** | 005056BCB79A1FD08FFEE8D5ABAB6665 (AB6665) |
| **Issue Name:** | TRM TRR Lack of Proactive Obsolescence Governance |
| **Action Driver Name:** | TRM TRR Lack of Proactive Obsolescence Governance |
| **Issue Source:** | Technology Risk Management (2LoD) – Technology Risk Review (TRR) finding. Evidence decks filed under reference 50427831 *[confirm TRR / issue number]* |
| **Scope:** | BSI datacenter technology obsolescence – 24 KRIs (aggregated, hardware, software and zero tolerance), approximately 840 software assets plus hardware lifecycle population, reported through KRISP. |
| **RCSA Linkage:** | BSI.IT.5.R4 – Obsolescence and Disposal (ID 50434969) |
| **Accountable Executive:** | Jose Oliveros *[Title – confirm]* |
| **Governance Owner:** | ITOPS – IT Obsolescence Management (Infrastructure owner: Ian Rojo) |
| **Action Driver Target Date:** | *[MM/DD/YYYY]* |
| **Test Period:** | July 2026 and August 2026 reporting cycles (IT Working Group meetings 8/21/2026 and 9/18/2026), with outcome confirmed against the 9/30/2026 KRISP evidence. |
| **Testing Performed by:** | IT RCSA Control Assessment Team |
| **Executive Summary** | IT RCSA performed validation testing to support closure of Action Driver 005056BCB79A1FD08FFEE8D5ABAB6665 (AB6665) – TRM TRR Lack of Proactive Obsolescence Governance. The TRM Technology Risk Review found that BSI IT lacked proactive governance of technology obsolescence; the Action Driver required BSI IT to move from reactive remediation of obsolete assets to a governed, forward-looking process: a formal governance forum, an 18–24 month forward view of end-of-life (EOL) exposure, per-asset remediation tracking, funding traceability and a defined escalation path.<br>IT RCSA inspected the obsolescence management materials presented to the BSI IT Working Group for the July and August 2026 reporting cycles and agreed the reported results to the 9/30/2026 KRISP evidence. All four test steps passed, subject to the disclosures noted below.<br>The process operated over two consecutive cycles and handled a real breach: when CI23514 (obsolete databases) moved to Red in July 2026, the issue was escalated, decommissions were committed and dependencies were tracked, and CI23514 returned to Green (9.259%) by the September and October 2026 submissions. Based on the test results, the TRR finding is considered Remediated and Action Driver AB6665 is recommended for closure, subject to TRM (2LoD) validation. |
| **Action Driver Description:** | Issue: BSI IT did not govern technology obsolescence proactively. Assets reached vendor end of life without a forward view, committed remediation dates or funding visibility, and dependencies on InfoSec, application owners and vendors were not consistently escalated. This contributed to sustained KRI breaches, notably CI23514 – Percentage of obsolete database instances.<br>Remediation objective: demonstrate that (1) a formal forum reviews obsolescence monthly with minutes, owners and dates; (2) an 18–24 month forward EOL view is rebuilt each cycle from KRISP asset-level data; (3) every planned asset is tracked with owner, action, target date and status; (4) funding is traceable each reporting period; and (5) dependencies and breaches are escalated and reported.<br>Root Cause (Level 1 / Level 2): *[Root cause per action driver record]*<br>Tools: KRISP (Value Submissions and evidence workbook), ServiceNow CMDB, ITOPS 24-month remediation register, BSI IT Working Group. |
| **Sampling Rationale:** | Governance is evidenced per monthly cycle. IT RCSA tested both IT Working Group cycles available after the remediation was implemented (July 2026 and August 2026 reporting periods) – the full population for the period – and confirmed the outcome against the subsequent 9/30/2026 KRISP evidence. |

## AD AB6665 Closure Validation – Testing Performed

**1.**

| Test Step: | 1 – Formal Governance Forum |
|---|---|
| **Attribute:** | Obtain evidence that obsolescence is reviewed monthly in a formal forum, with minutes recording named owners and dates, and that results are reported to senior IT management. |
| **Population / Sample:** | Two monthly cycles: July 2026 (IT WG 8/21/2026) and August 2026 (IT WG 9/18/2026). |
| **Business Artifacts:** | • AB6665.01 IT Working Group – August 2026 (obsolescence section, slides 63–69)<br>• AB6665.02 IT Working Group – September 2026 (obsolescence section, slides 60–67) |
| **Review:** | • A dedicated "IT Obsolescence Management" section was presented to the BSI IT Working Group in both cycles: executive summary, portfolio dashboard, roadmap, KRI detail and forward EOL view.<br>• August 2026 deck records the governance control "Formal governance forum – monthly ITOPS obsolescence review running; 6 Aug 2026 minutes filed with named owners and dates."<br>• Both decks report status, escalations and planned actions for the cycle; the IT Working Group includes the Accountable Executive and IT domain owners. |
| **Result:** | **Pass** |

**2.**

| Test Step: | 2 – 18–24 Month Forward EOL View |
|---|---|
| **Attribute:** | Validate that a forward-looking view of EOL exposure covering at least 18–24 months is produced each cycle from KRISP asset-level dates. |
| **Population / Sample:** | Forward EOL view in each of the two cycles. |
| **Business Artifacts:** | • AB6665.01 slide 69 – 24-Month Forward-Looking EOL View (July 2026)<br>• AB6665.02 slide 67 – 24-Month Forward-Looking EOL View (August 2026) |
| **Review:** | • Horizon rebuilt each month, derived from KRISP asset-level obsolescence dates and de-duplicated by CI ID.<br>• July 2026 (through July 2028): 35 already EOL, 68 due ≤6 months, 27 in 7–12 months, 53 in 13–24 months – 183 software assets (+57 hardware).<br>• August 2026 (through August 2028): 27 already EOL, 59 due ≤6 months, 28 in 7–12 months, 52 in 13–24 months – 166 software assets (+59 hardware).<br>• Next wave identified in advance: 59 Windows Server 2016 / IIS 10 assets reaching vendor EOL on 12 Jan 2027. |
| **Result:** | **Pass** |

**3.**

| Test Step: | 3 – Per-Asset Tracking and Funding |
|---|---|
| **Attribute:** | Validate that planned assets are tracked with owner, action, target date and status, and that the funding position is stated each reporting period. |
| **Population / Sample:** | Remediation roadmap in each of the two cycles. |
| **Business Artifacts:** | • AB6665.01 slide 65 – Remediation Roadmap (240 assets)<br>• AB6665.02 slides 61 and 63 – Governance controls; Remediation Roadmap (225 assets) |
| **Review:** | • August 2026 deck records "Per-asset remediation tracking – register of the 225 planned assets with owner, action, target date and status."<br>• Roadmap allocates every planned asset to a remediation lane and half-year (H2 2026: 32, H1 2027: 84, H2 2027: 55, H1 2028: 54 = 225), with net change vs the prior plan (–57).<br>• Funding stated on the roadmap in both cycles: "approved IT budget · additional requested this period: none · pending approval: none." |
| **Result:** | **Pass** |

**4.**

| Test Step: | 4 – Escalation and Outcome |
|---|---|
| **Attribute:** | Validate that breaches and remediation dependencies are escalated and reported every cycle and that the process produces measurable improvement. |
| **Population / Sample:** | Escalations reported in both cycles; outcome agreed to 9/30/2026 KRISP evidence. |
| **Business Artifacts:** | • AB6665.01 / AB6665.02 executive summaries<br>• AB6665.03 09.30.2026 Miami Obsolescence KRISP evidence v3.xlsx (TWG Metrics)<br>• AB6665.04 KRISP Value Submissions screenshot – CI23514 |
| **Review:** | • July cycle: CI23514 Red (19.35%) escalated – 7 instances reached vendor EOL 14 Jul; 4 August decommissions committed; Varonis and CyberArk database decommissions escalated to InfoSec.<br>• August cycle: dependencies recorded – Varonis (InfoSec decommission date), Tableau PostgreSQL and xmfax MySQL (application owner / vendor dates).<br>• Outcome: obsolete software assets 35 (4.13%) July → 27 (3.22%) August → 20 (AOB0001.D 2.378%, Green) at 9/30/2026.<br>• CI23514 Red (Aug 2026 submission) → Green 9.259% (Sep and Oct 2026 submissions); all 9 zero tolerance KRIs at 0.00% throughout. |
| **Result:** | **Pass** |

| Obsolescence Outcome by Reporting Cycle | | | |
|---|---|---|---|
| **Measure** | **July 2026 (as of 7/31)** | **August 2026 (as of 8/31)** | **September 2026 (as of 9/30)** |
| **Obsolete software assets (AOB0001.D)** | 35 / 847 (4.13%) | 27 / 838 (3.22%) | 20 / 841 (2.378%) |
| **KRIs Green (of 24)** | 23 (1 Red: CI23514) | 23 (1 Amber per ITOPS deck: CI23514) | 24 (CI23514 Green) |
| **CI23514 – obsolete database instances** | 12 / 62 (19.355%) Red | 5 / 54 (9.259%) Green (KRISP) | 5 / 54 (9.259%) Green |
| **Zero tolerance KRIs at 0.00%** | 9 / 9 | 9 / 9 | 9 / 9 |
| **Assets in 24-month plan** | 240 (183 SW + 57 HW) | 225 (166 SW + 59 HW) | n/a – next IT WG cycle |

*Sources: IT Working Group decks (AB6665.01 / .02), KRISP evidence 9/30/2026 (AB6665.03), KRISP screenshot (AB6665.04).*

| 24-Month Remediation Roadmap – August 2026 Cycle (assets reaching EOL per period) | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Lane** | **Action** | **H2 2026** | **H1 2027** | **H2 2027** | **H1 2028** | **Total** | **vs June** |
| **Windows Servers** | In-place OS upgrade / rebuild | 6 | 39 | – | – | 45 | −21 |
| **Linux Servers** | In-place OS upgrade | 6 | – | – | – | 6 | −24 |
| **Web Servers** | IIS upgrade / app migration | 5 | 16 | – | – | 21 | −6 |
| **Databases** | Engine upgrade / migration | 8 | 4 | 6 | – | 18 | −4 |
| **Hypervisors** | ESXi upgrade (accelerated to H2 2026) | – | – | 31 | – | 31 | +1 |
| **Network Devices** | Firmware / HW refresh | 2 | 13 | 8 | – | 23 | −2 |
| **App Servers (WAS)** | Runtime upgrade / migration | – | 12 | 8 | – | 20 | 0 |
| **Physical Servers** | Hardware replacement | 3 | – | – | 48 | 51 | −1 |
| **Storage** | Controller / array refresh | 2 | – | 2 | 6 | 10 | 0 |
| **Total assets in plan** | | **32** | **84** | **55** | **54** | **225** | **−57** |

*Funding: approved IT budget · Additional requested this period: none · Pending approval: none. Source: AB6665.02 slide 63.*

| Points Noted During Testing | |
|---|---|
| **Disclosures:** | • The ITOPS monthly obsolescence review minutes (6 Aug 2026) and the 225-asset remediation register are referenced in the IT Working Group deck but were not provided with this package. *[Obtain and attach as AB6665.05 / AB6665.06 if 2LoD/TRM or IA require direct evidence.]*<br>• Two operating cycles were tested (July and August 2026). The September 2026 cycle (IT Working Group October 2026) was not available; the 9/30/2026 KRISP evidence confirms the outcome trend continued.<br>• Reporting inconsistency: the September 18, 2026 deck reports CI23514 for August 2026 as Amber at 14.04% (8 / 57), while the approved KRISP September 2026 submission (as of 8/31/2026) and the KRI table in the same deck report 9.259% (5 / 54) Green. *[Obtain management explanation – e.g., snapshot taken before CMDB record clean-up.]* Consistency between the ITOPS deck and KRISP should be checked each cycle.<br>• Open dependencies remain outside ITOPS control – Varonis databases (InfoSec decommission date), Tableau PostgreSQL and xmfax MySQL (application owner / vendor dates). They are recorded and reported every cycle, which is the governance expectation; remediation of the assets themselves continues under the roadmap.<br>• The August 2026 IT Working Group KRI commentary listed "proactive obsolescence governance" among past-due BSI Technology action drivers pending validation. *[Confirm current AD due date; TRM (2LoD) validation of this TRR finding is expected before closure.]* |

| Closure Package – Artifact Index | | |
|---|---|---|
| **Reference** | **Artifact** | **Supports** |
| **AB6665.01** | IT Working Group - August 2026 2 (50427831).pptx – reporting period July 2026, meeting 8/21/2026 | Forum, forward EOL view, roadmap, July escalations (steps 1–4) |
| **AB6665.02** | IT Working Group - September 20269 (50427831).pptx – reporting period August 2026, meeting 9/18/2026 | Governance controls, per-asset register, funding, escalations, progress (steps 1–4) |
| **AB6665.03** | 09.30.2026 Miami Obsolescence KRISP evidence v3.xlsx (TWG Metrics) – attached to "BSI Miami - September KRISP metrics evidence file" email, 10/02/2026 | Outcome at 9/30/2026: AOB0001.D 2.378%, CI23514 9.259%, zero tolerance 0.00% (step 4) |
| **AB6665.04** | KRISP Value Submissions screenshot – CI23514 history (captured 10/19/2026) | CI23514 Red → Green after escalation (step 4) |

| | | |
|---|---|---|
| **Related** | BSI.IT.5.R4 CI23514 / AD 36BA1 Closure Package (separate) | Closure of the CI23514 KRI breach AD 005056BCA24E1FD0AEE7F6D7CD136BA1 |

| Validation Results | |
|---|---|
| **Results** | **Justification** |
| **Overview of Remediation Timeline/Status** | |
| **Overall Target Date:** | *[MM/DD/YYYY]* |
| **Remediation Status:** | Remediated |
| **Control Performance Rating:** | Satisfactory |
| **Overall Validation Remediation Conclusion** | |
| IT RCSA reviewed Action Driver 005056BCB79A1FD08FFEE8D5ABAB6665 (AB6665) – TRM TRR Lack of Proactive Obsolescence Governance – including the IT Obsolescence Management materials presented to the BSI IT Working Group for the July and August 2026 reporting cycles and the 9/30/2026 KRISP evidence. Testing confirmed that obsolescence is reviewed monthly in a formal forum with minutes, owners and dates; that a 24-month forward EOL view is rebuilt each cycle from KRISP asset-level dates; that planned assets are tracked in a per-asset register and allocated to a dated remediation roadmap with the funding position stated; and that breaches and dependencies are escalated and reported each cycle. The process produced measurable improvement – obsolete software assets fell from 35 to 20 and CI23514 returned from Red to Green. Based on the test results the control performance is rated Satisfactory, and Action Driver 005056BCB79A1FD08FFEE8D5ABAB6665 (AB6665) is considered Remediated, subject to resolution of the disclosures noted above. | |
| **Status:** | *[Validation Complete. No further action needed.]* |
| **Completed by:** | *[Name]* |
| **Validation Date:** | *[MM/DD/YYYY]* |

| Manager Review | |
|---|---|
| **Status:** | *[Validation Complete. No further action needed.]* |
| **Completed by:** | *[Name]* |
| **Validation Date:** | *[MM/DD/YYYY]* |

## Progress at a Glance

| 35 → 27 | 4.05% → 3.22% | 282 → 225 | 9 / 9 |
|---|---|---|---|
| Obsolete assets<br>7 fewer | Obsolescence rate (AOB0001.D)<br>Green, well inside threshold | Assets in the 24-month plan<br>57 fewer to remediate | Zero Tolerance KRIs Green<br>0.00% in every cycle, no breach |

**Remediation delivered, June to August**

- **Databases back from Red to Amber**
  Obsolete instances 12 (July) → 8. BigFix and SW databases migrated, miantsql decommissioned.
- **Windows Servers**
  Obsolete 11 → 6, Current 125 → 141 as Windows 2016 upgrades land ahead of the Jan 2027 EOL.
- **Linux Servers**
  Outgoing 26 → 0 and Current 91 → 115; the 6 obsolete hosts that remain are all dated for remediation.
- **Physical Servers**
  The 13 hosts under validation in June are cleared; 3 remain, all under maintenance until Dec 2030.
- Network Devices
  Obsolete 4 → 2 (ash-dr-nk5 pair decommissioned); 5K core refresh in execution, completion 30 Sep 2026.
- **Hypervisors**
  0 obsolete; all 31 ESXi hosts in Outgoing ahead of the Oct 2027 EOL.
- **CMDB hygiene**
  Retired records cleaned up and ash-dr-voice moved to its 2028 EOL, so KRISP now reflects real risk.

**Governance controls in place (audit finding coverage)**

- **Formal governance forum**
  Monthly ITOPS obsolescence review running; 6 Aug 2026 minutes filed with named owners and dates.
- **18–24 month forward view**
  24-month EOL horizon rebuilt every month from KRISP asset-level dates (Slide 9).
- **Per-asset remediation tracking**
  Register of the 225 planned assets with owner, action, target date and status.
- **Funding traceability**
  Budget position stated on the roadmap each reporting period (Slide 5).
- **Escalation path**
  InfoSec, application owner and vendor dependencies recorded and reported every cycle.

**Obsolete assets by reporting cycle** → 35 (4.13%) July 2026 → 27 (3.22%) August 2026

IT Obsolescence Management | Confidential | Reporting Period: August 2026 — Santander — 3

*Figure A1 – IT Working Group 9/18/2026, "Progress at a Glance": governance controls in place (forum, forward view, per-asset tracking, funding, escalation) (AB6665.02)*

## Remediation Roadmap: 24-Month Delivery Plan

225 assets in the plan through Aug 2028 (166 software + 59 hardware) | ITOPS-owned | Cell values = assets reaching end of life in that period, from KRISP dates, subject to confirmation

| 27 | 3.22% | 225 | 32 |
|---|---|---|---|
| Obsolete assets today<br>34 in June (net 7 fewer) | Obsolescence rate<br>4.05% in June, within threshold | Assets in 24-month plan<br>282 in June (57 fewer) | Past EOL, in remediation now<br>27 software + 5 hardware |

| Remediation lane | H2 2026 (now)<br>past EOL + accelerated | H1 2027<br>Jan–Jun 2027 | H2 2027<br>Jul–Dec 2027 | H1 2028<br>Jan–Jun 2028 | Total<br>in plan | vs June plan<br>net change |
|---|---|---|---|---|---|---|
| Windows Servers<br>In-place OS upgrade / rebuild | 6 | 39 | – | – | 45 | −21 |
| Linux Servers<br>In-place OS upgrade | 6 | – | – | – | 6 | −24 |
| Web Servers<br>IIS upgrade / app migration | 5 | 16 | – | – | 21 | −6 |
| Databases<br>Engine upgrade / migration | 8 | 4 | 6 | – | 18 | −4 |
| Hypervisors | | – | 3 1 | – | 31 | +1 |
| Network Devices<br>Firmware / HW refresh | 2 | 13 | 8 | – | 23 | −2 |
| App Servers (WAS)<br>Runtime upgrade / migration | – | 12 | 8 | – | 20 | 0 |
| Physical Servers<br>Hardware replacement | 3 | – | – | 48 | 51 | −1 |
| Storage<br>Controller / array refresh | 2 | – | 2 | 6 | 10 | 0 |
| **Total assets in plan** | **32** | **84** | **55** | **54** | **225** | **−57** |

**What ITOPS remediates in each period**

*Funding: approved IT budget · Additional requested this period: none · Pending approval: none*

**H2 2026 (now)**
27 software + 5 hardware assets already past EOL, plus 31 ESXi
**32 assets**

**H1 2027**
Windows Server 2016 and IIS wave (EOL 12 Jan 2027), then WAS, network and database items Mar–May 2027
**84 assets**

**H2 2027**
Databases, WAS and network devices due Jul–Dec 2027 (55 due; 31 hypervisors already moved to H2 2026)
**55assets**

**H1 2028**
Hardware refresh wave: 48 physical servers and 6 storage arrays reaching hardware lifecycle end
**54 assets**

IT Obsolescence Management | Confidential | Reporting Period: August 2026 — Santander — 5

*Figure A2 – IT Working Group 9/18/2026, "Remediation Roadmap: 24-Month Delivery Plan" with funding position (AB6665.02)*

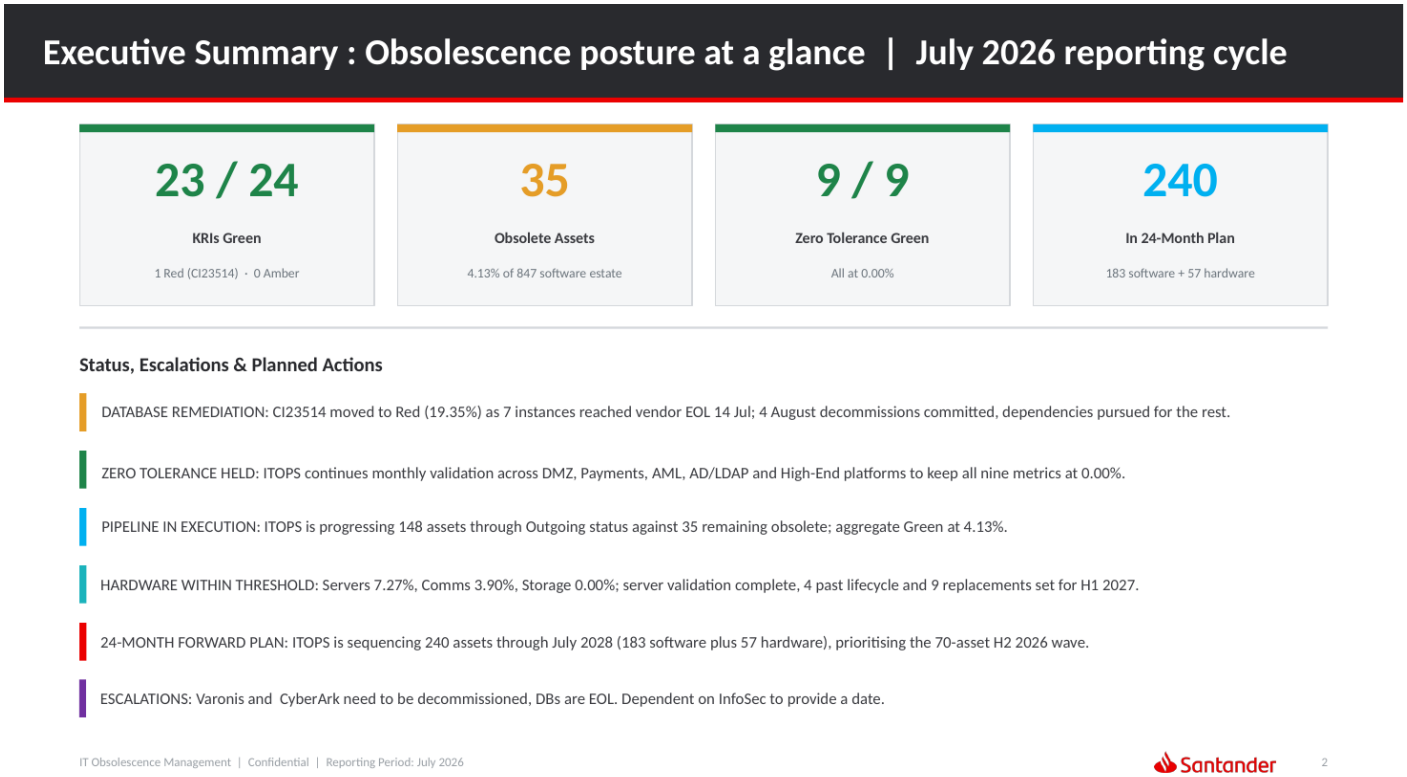



*Figure A3 – IT Working Group 8/21/2026, Executive Summary for July 2026 cycle: CI23514 Red escalation and InfoSec dependencies (AB6665.01)*